

Ah-Yeong Jeong

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RESEARCH INTERESTS

Brain Dynamics Modeling with State-Space Models, Brain Signal Representation Learning (fMRI, EEG), Foundation Models for Brain Signals, Brain Disorder Diagnosis

EDUCATION

Korea University

Mar 2025 – Present

- M.S. in Artificial Intelligence (expected Aug 2027)
- Advised by Professor [Heung-Il Suk](#) at Machine Intelligence Lab. ([MILab](#))

Korea University

Mar 2020 – Feb 2025

- B.S. in Biomedical Engineering, with an Interdisciplinary Major in Brain and Cognitive Sciences
- **Selected Coursework:** Biomedical AI, Machine Learning, Biomedical Signals and Systems, Digital Signal Processing, Biomedical Statistics

PUBLICATIONS

Under Review

[1] CHORD: A State-Space EEG Foundation Model with Low-Rank Cross-Channel Coupling

Ah-Yeong Jeong, Heung-Il Suk
Under review, 2026

[2] A Unified Spatiotemporal SSM for Dynamic Brain State Analysis in rs-fMRI

Ah-Yeong Jeong, Seungwoo Jeong, Da-Woon Heo, Heung-Il Suk
Under review, 2026

Peer-Reviewed Publications

[3] Complex wavelet-based Transformer for neurodevelopmental disorder diagnosis via direct modeling of real and imaginary components [\[Paper\]](#)

Ah-Yeong Jeong*, Da-Woon Heo*, Heung-Il Suk
Medical Image Analysis, vol. 109, 2026

[4] BrainWaveNet: Wavelet-based Transformer for Autism Spectrum Disorder Diagnosis [\[Paper\]](#)

Ah-Yeong Jeong*, Da-Woon Heo*, Eunsong Kang, Heung-Il Suk
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024 (**Oral Presentation, Best Paper Award Shortlist**)
(* denotes equal contribution)

RESEARCH EXPERIENCE

Machine Intelligence Lab, [MILab](#) | Korea University

Nov 2023 – Feb 2025

- Undergraduate Research Intern
- Advised by Professor [Heung-Il Suk](#), Korea University

K-NeuroMind NEXUS | National R&D Project (NRF/MSIT, Korea)

Apr 2026 – Present

- Neurocognitive-Affective Next-Generation Cross-modal Foundation Platform, a multi-institutional project of Ewha Womans University (lead), Korea University, Seoul National University, and Hanyang University

- Led the writing of the funded research proposal for the Korea University team (PI: Prof. [Heung-Il Suk](#))
- Currently developing an EEG foundation model as a core researcher on the project

SELECTED PROJECTS

Diffusion Models for Longitudinal MRI Synthesis and Brain Tumor Growth Prediction |

Graduate Course Project

Fall 2025

- Developed diffusion-based generative models that synthesize follow-up MRI scans and predict brain tumor growth from longitudinal imaging
- Course project for Computer Vision Application and Practice, built on the [Yale-Brain-Mets-Longitudinal](#) (TCIA) and [SAILOR](#) (EBRAINS) longitudinal oncology cohorts

PATENTS & SOFTWARE COPYRIGHTS

Patent

- Method and apparatus for assisting diagnosis of brain diseases based on complex wavelet transform and integrated spectral-spatiotemporal modeling (Korean Patent Application No. 10-2026-0136283)

Software Copyright Registrations

- Real-imaginary cross-attention Transformer program for neurodevelopmental disorder classification from functional brain imaging (No. C-2026-035214)
- Spatiotemporal state-space model program for dynamic brain state analysis (No. C-2026-035215)
- Wavelet-based Transformer software for autism spectrum disorder classification from functional MRI (No. C-2026-035216)

ACADEMIC SERVICE

Reviewer: Medical Image Analysis (2025), AAAI (2026)

VOLUNTEERING

Korea University Brain Science Academic Society, NewLearn

Mar 2022 - Dec 2023

- Vice President (2023)
- Advised by Professor [Dong-Joo Kim](#)
- Organized and led interdisciplinary academic activities across neuroscience, cognitive science, and mathematics
- Attended KSBNS (Korean Society for Brain and Neural Sciences) 2022

Computational Neuroscience academic club, Neuromatch

Mar 2022 - Dec 2023

- Team Leader (2023)
- Organized group study and discussions on computational neuroscience
- Completed [Neuromatch Academy](#) (NMA) 2022

National Union of College Students in Brain and Cognitive Sciences, Synapse *Jun 2022 - Dec 2023*

- Member
- Organized inter-university symposiums across brain science societies

SKILLS

- **Languages:** Korean (Native), English
- **Programming:** Python (PyTorch, TensorFlow), MATLAB, Docker, Bash/Shell
- **Neuroimaging Tools:** DPABI, C-PAC (fMRI preprocessing pipelines)